Course: Cloud and Network Security CNS-C2-2026

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Week 5 Assignment 2:
Configuring Site-to-Site VPNs

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Objectives

- Verify connectivity throughout the network.
- Configure R1 to support a site-to-site IPsec VPN with R3.

Introduction

In this lab, I configured a site-to-site IPsec VPN between routers R1 and R3 to securely encrypt traffic between their LANs. I set up the security policies, pre-shared key, encryption settings, access control lists, and crypto maps on both routers to establish the VPN tunnel, while R2 simply forwarded traffic without requiring any VPN configuration. I then tested the connection by pinging between the end devices and checking the IPsec security association counters to confirm that traffic between PC-A and PC-C was encrypted, while traffic outside the VPN, such as communication with PC-B, remained unencrypted. The lab demonstrated how IPsec secures communication between two networks over an untrusted network.

Instructions

Part 1: Configure IPsec Parameters on R1

- **Step 1: Test connectivity.**
- **Step 2: Enable the Security Technology package.**
- **Step 3: Identify interesting traffic on R1.**
- **Step 4: Configure the IKE Phase 1 ISAKMP policy on R1.**
- **Step 5: Configure the IKE Phase 2 IPsec policy on R1.**
- **Step 6: Configure the crypto map on the outgoing interface.**

Part 2: Configure IPsec Parameters on R3

- **Step 1: Enable the Security Technology package.**
- **Step 2: Configure router R3 to support a site-to-site VPN with R1.**
- **Step 3: Configure the IKE Phase 1 ISAKMP properties on R3.**
- **Step 4: Configure the IKE Phase 2 IPsec policy on R3.**
- **Step 5: Configure the crypto map on the outgoing interface.**

Part 3: Verify the IPsec VPN

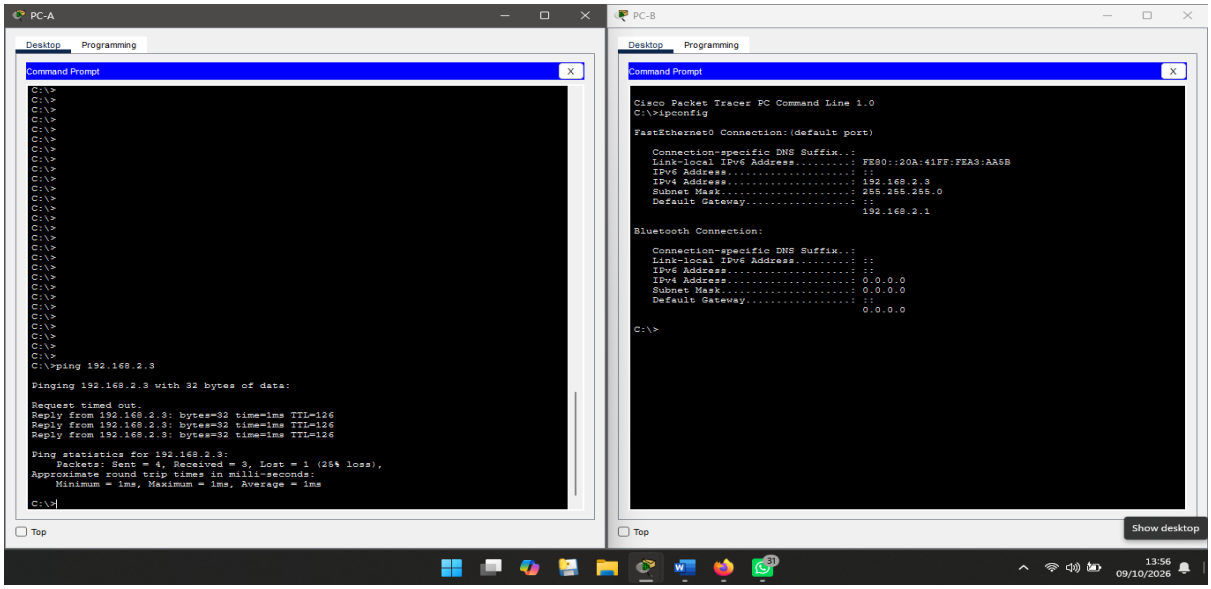
- **Step 1: Verify the tunnel prior to interesting traffic.**
- **Step 2: Create interesting traffic.**
- **Step 3: Verify the tunnel after interesting traffic.**
- **Step 4: Create uninteresting traffic.**

- **Step 5: Verify the tunnel.**.
- **Step 6: Check results.**

Part 1: Configure IPsec Parameters on R1

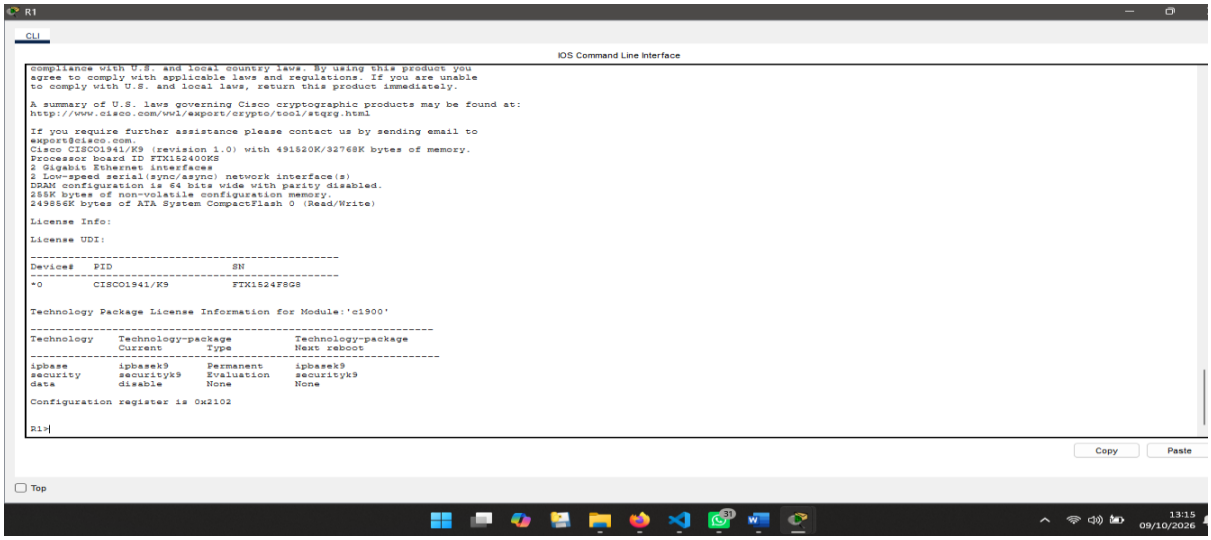
Step 1: Test connectivity.

Ping from PC-A to PC-C.



Step 2: Enable the Security Technology package.

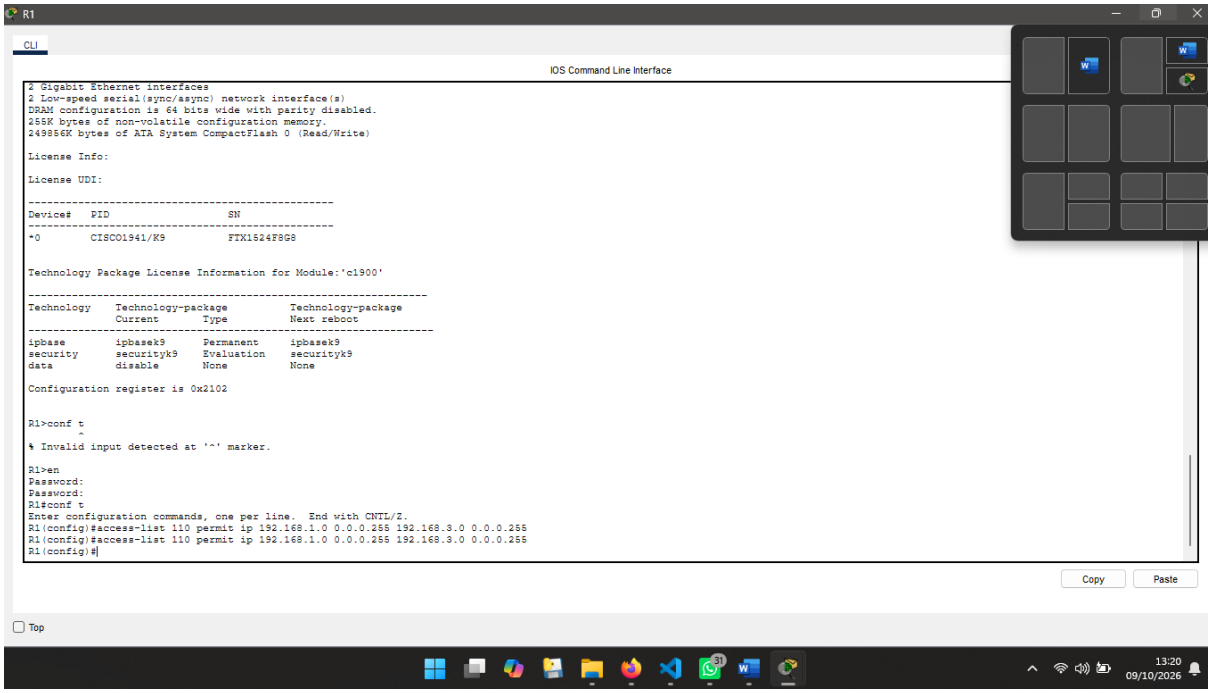
Enable the security technology package by using the following command to enable the package. R1(config)# **license boot module c1900 technology-package securityk9.** Accept the end-user license agreement. Save the running-config and reload the router to enable the security license. Verify that the Security Technology package has been enabled by using the **show version** command.



Step 3: Identify interesting traffic on R1.

Configure ACL 110 to identify the traffic from the LAN on R1 to the LAN on R3 as interesting. This interesting traffic will trigger the IPsec VPN to be implemented when there is traffic between the R1 to R3 LANs. All other traffic sourced from the LANs will not be encrypted. Because of the implicit **deny all**, there is no need to configure a **deny ip any any** statement.

R1(config)# **access-list 110 permit ip 192.168.1.0 0.0.0.255 192.168.3.0 0.0.0.255**



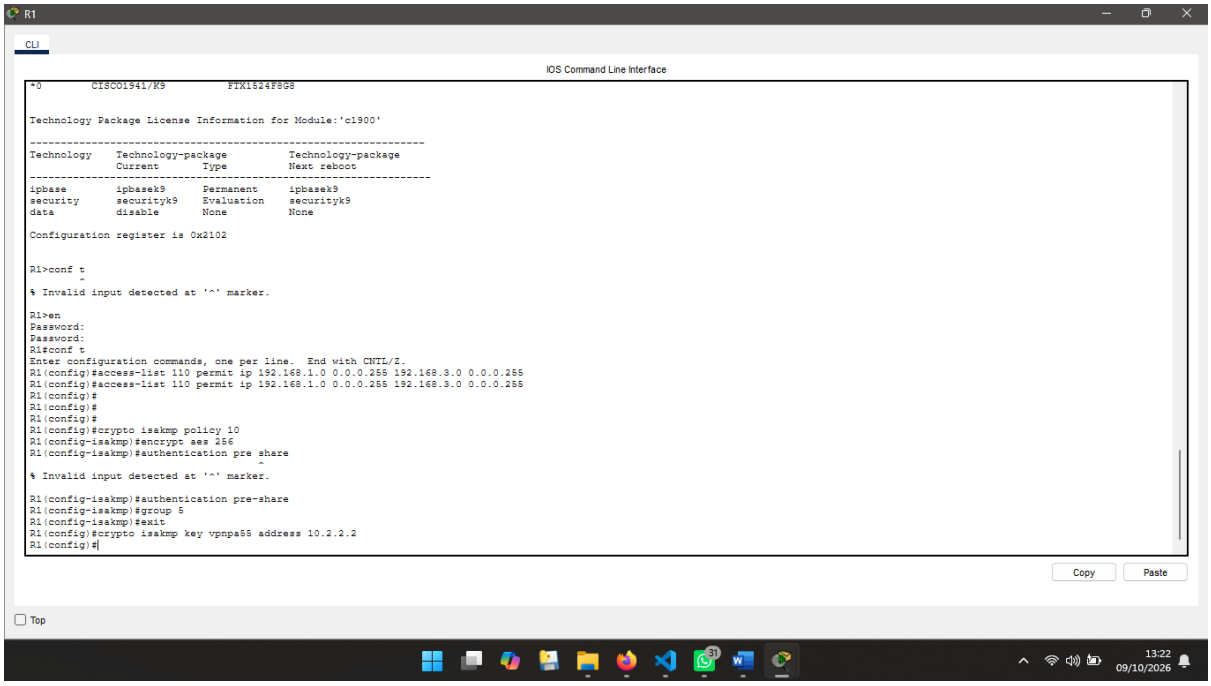
Step 4: Configure the IKE Phase 1 ISAKMP policy on R1.

Configure the **crypto ISAKMP policy 10** properties on R1 along with the shared crypto key **vpnpa55**. Refer to the ISAKMP Phase 1 table for the specific parameters to configure. Default values do not have to be configured. Therefore, only the encryption method, key exchange method, and DH method must be configured.

Note: The highest DH group currently supported by Packet Tracer is group 5. In a production network, you would configure at least DH 14.

R1(config)# **crypto isakmp policy 10,** R1(config-isakmp)# **encryption aes 256,** R1(config-isakmp)# **authentication pre-share,** R1(config-isakmp)# **group 5,**R1(config-isakmp)# **exit**

R1(config)# **crypto isakmp key vpnpa55 address 10.2.2.2**



Step 5: Configure the IKE Phase 2 IPsec policy on R1.

a. Create the transform-set VPN-SET to use **esp-aes** and **esp-sha-hmac**.

R1(config)# **crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac**

b. Create the crypto map VPN-MAP that binds all of the Phase 2 parameters together. Use sequence number 10 and identify it as an ipsec-isakmp map.

R1(config)# **crypto map VPN-MAP 10 ipsec-isakmp**

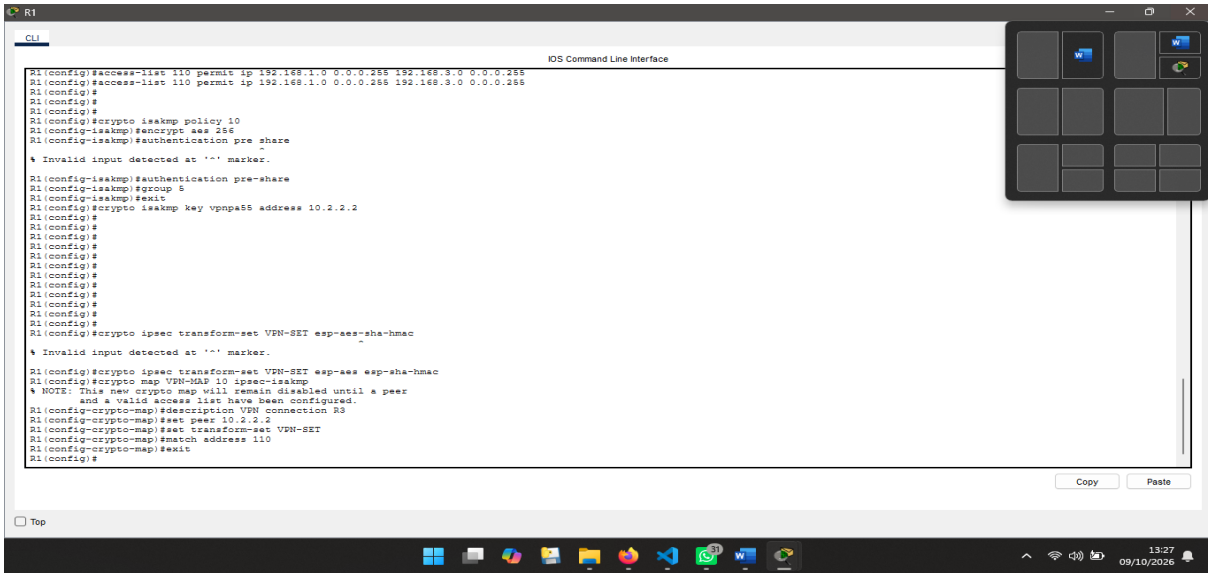
R1(config-crypto-map)# **description VPN connection to R3**

R1(config-crypto-map)# **set peer 10.2.2.2**

R1(config-crypto-map)# **set transform-set VPN-SET**

R1(config-crypto-map)# **match address 110**

R1(config-crypto-map)# **exit**

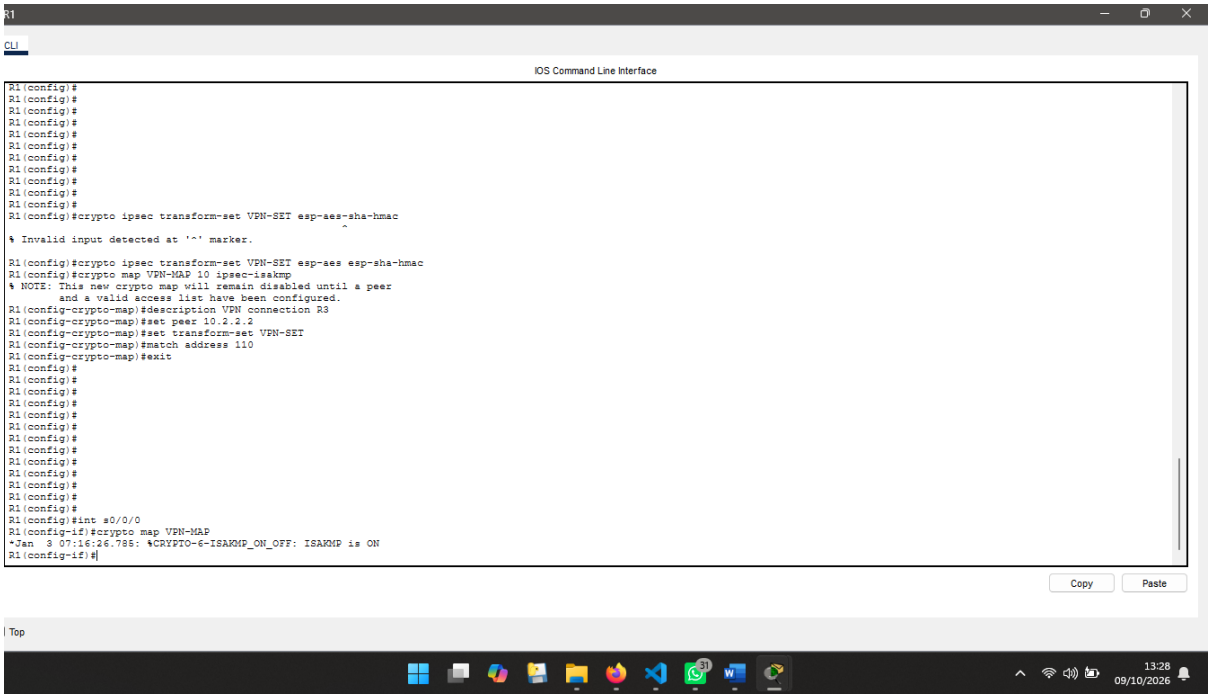


Step 6: Configure the crypto map on the outgoing interface.

Bind the **VPN-MAP** crypto map to the outgoing Serial 0/0/0 interface.

R1(config)# **interface s0/0/0**

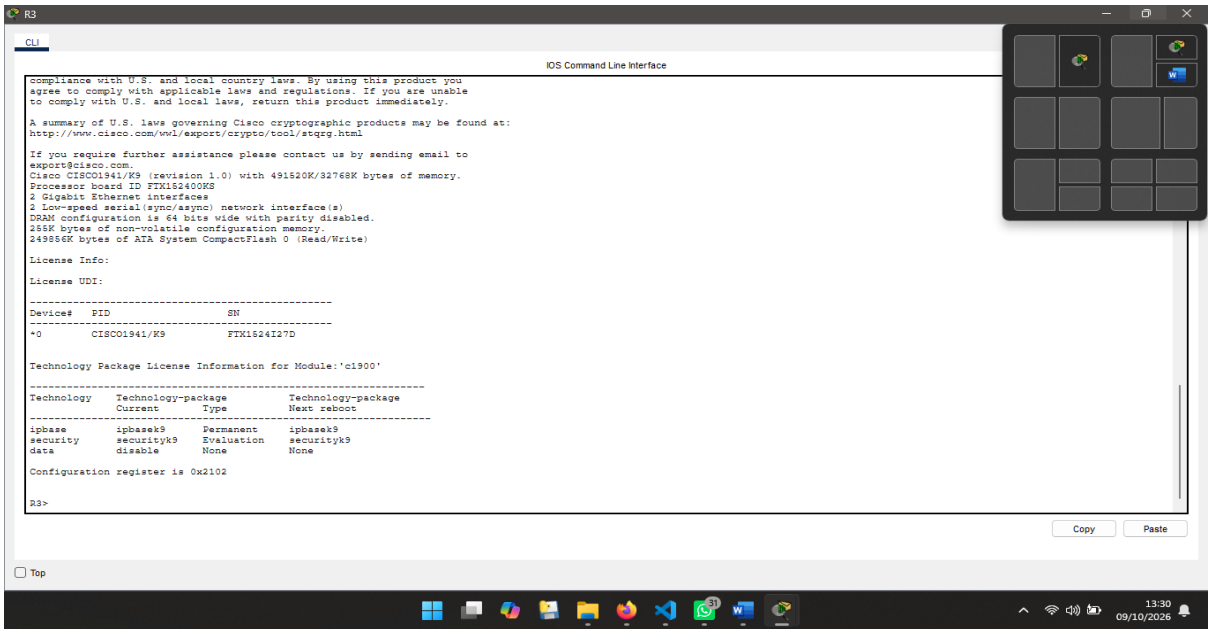
R1(config-if)# **crypto map VPN-MAP**



Part 2: Configure IPsec Parameters on R3

Step 1: Enable the Security Technology package.

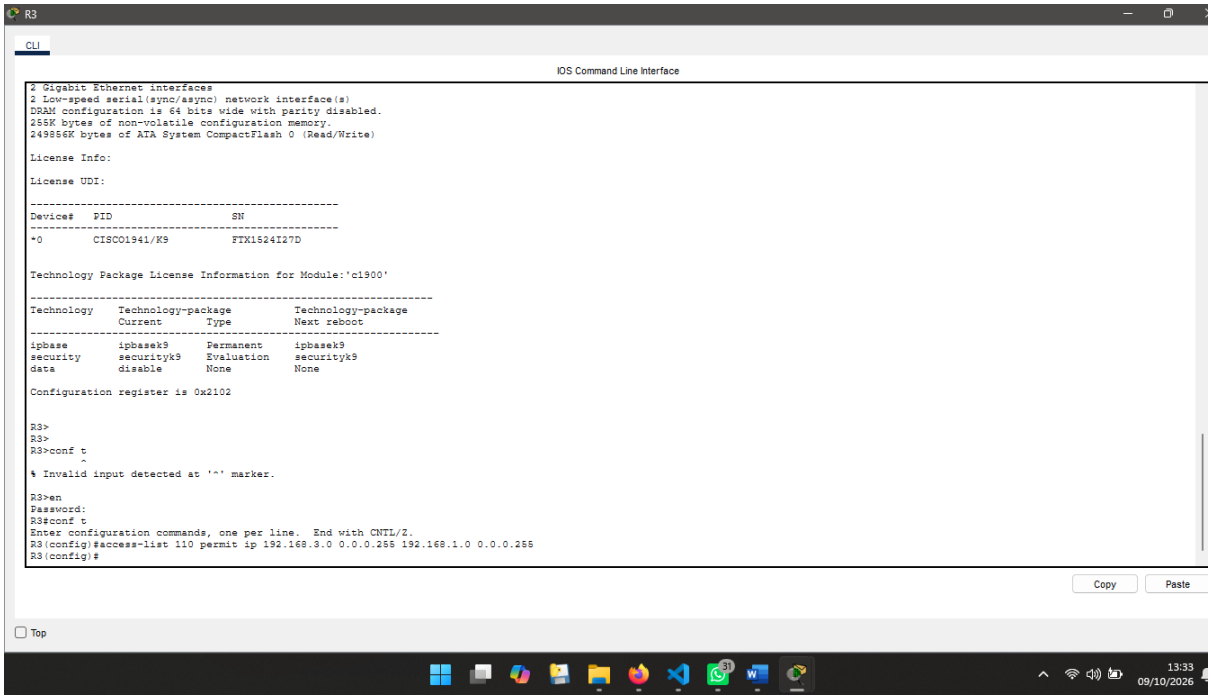
a. On R3, issue the **show version** command to verify that the Security Technology package license information has been enabled.



Step 2: Configure router R3 to support a site-to-site VPN with R1.

Configure reciprocating parameters on R3. Configure ACL 110 to identify the traffic from the LAN on R3 to the LAN on R1 as interesting.

R3(config)# **access-list 110 permit ip 192.168.3.0 0.0.0.255 192.168.1.0 0.0.0.255**

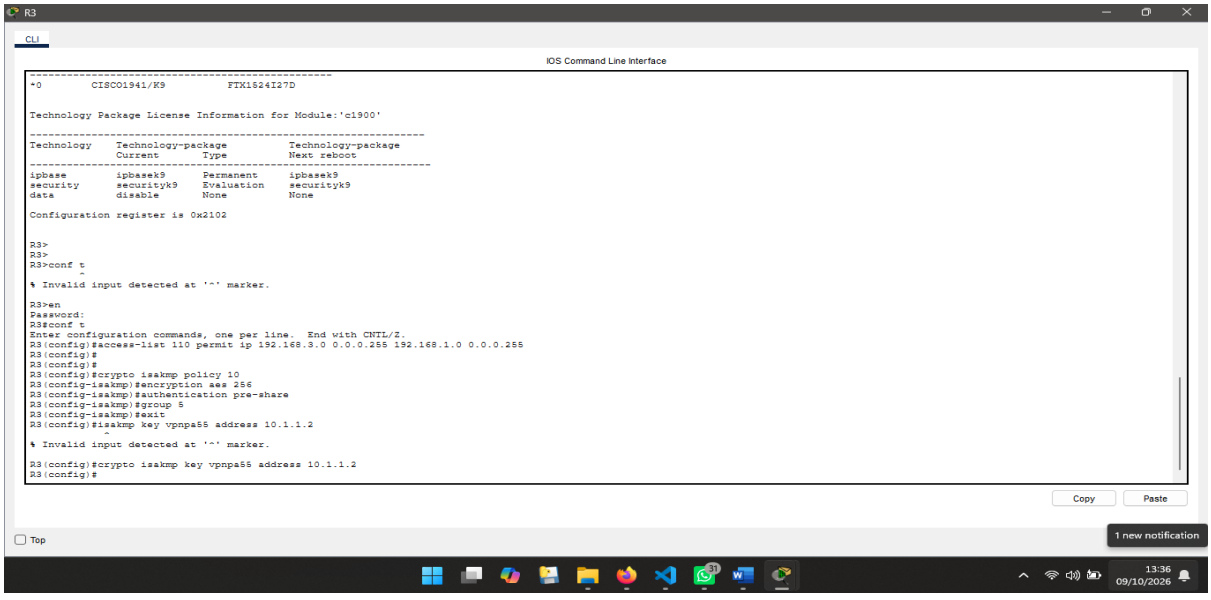


Step 3: Configure the IKE Phase 1 ISAKMP properties on R3.

Configure the crypto ISAKMP policy 10 properties on R3 along with the shared crypto key **vpnpa55**.

R3(config)# **crypto isakmp policy 10,** R3(config-isakmp)# **encryption aes 256,** R3(config-isakmp)# **authentication pre-share,** R3(config-isakmp)# **group 5,** R3(config-isakmp)# **exit**

R3(config)# **crypto isakmp key vpnpa55 address 10.1.1.2**



Step 4: Configure the IKE Phase 2 IPsec policy on R3.

c. Create the transform-set VPN-SET to use **esp-aes** and **esp-sha-hmac**.

R3(config)# **crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac**

d. Create the crypto map VPN-MAP to bind all of the Phase 2 parameters together. Use sequence number 10 and identify it as an ipsec-isakmp map.

R3(config)# **crypto map VPN-MAP 10 ipsec-isakmp**

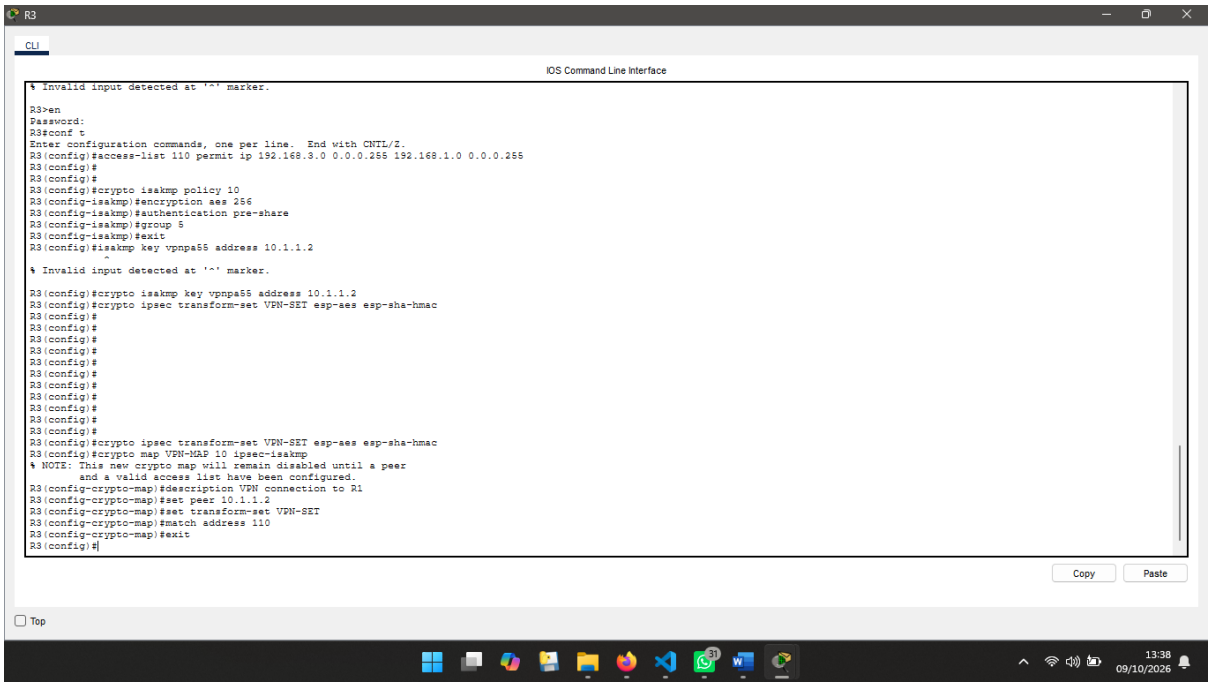
R3(config-crypto-map)# **description VPN connection to R1**

R3(config-crypto-map)# **set peer 10.1.1.2**

R3(config-crypto-map)# **set transform-set VPN-SET**

R3(config-crypto-map)# **match address 110**

R3(config-crypto-map)# **exit**

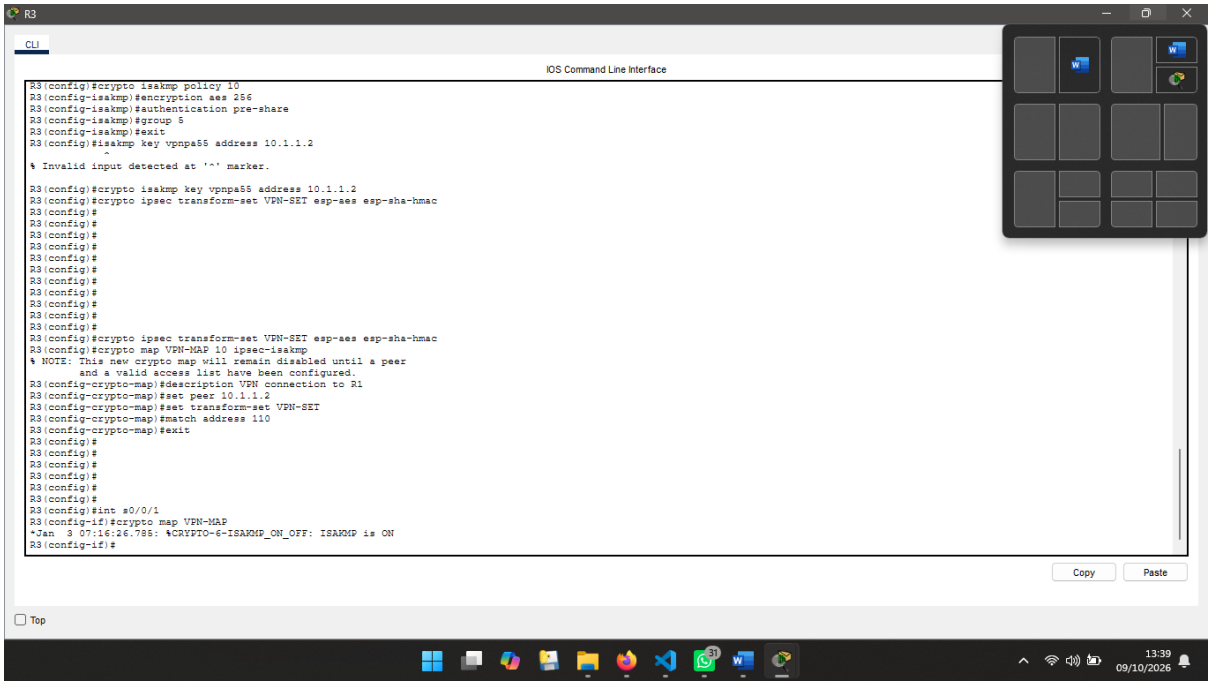


Step 5: Configure the crypto map on the outgoing interface.

Bind the VPN-MAP crypto map to the outgoing Serial 0/0/1 interface.

R3(config)# **interface s0/0/1**

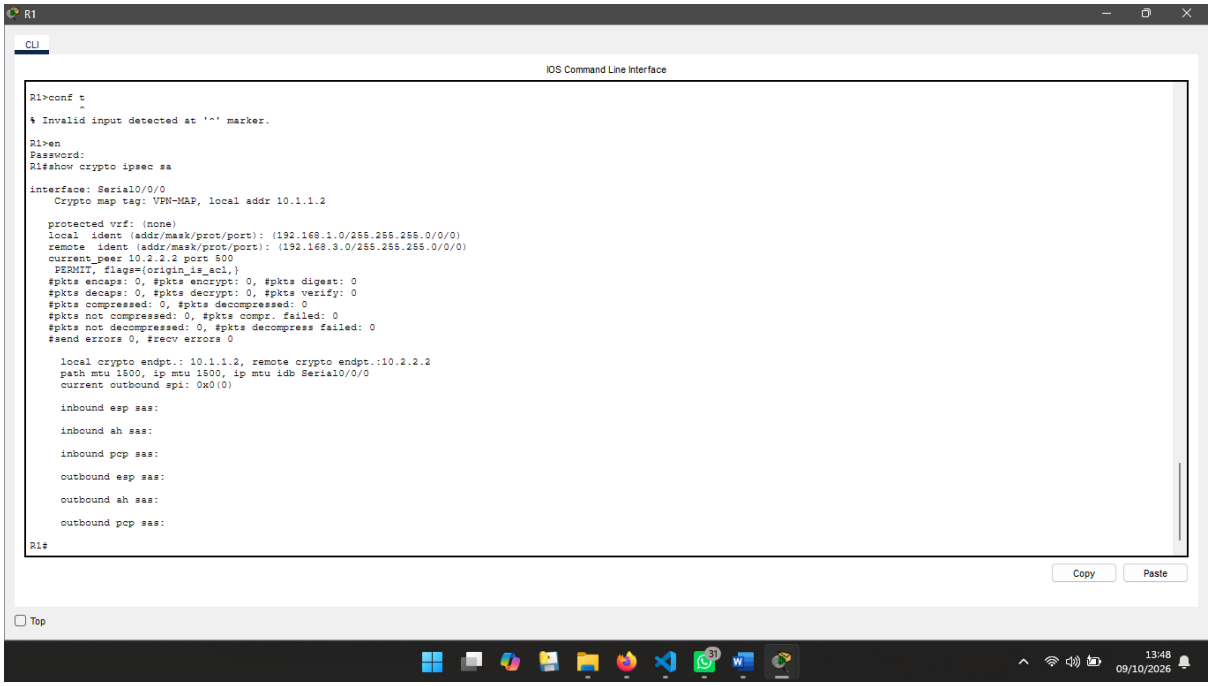
R3(config-if)# **crypto map VPN-MAP**



Part 3: Verify the IPsec VPN

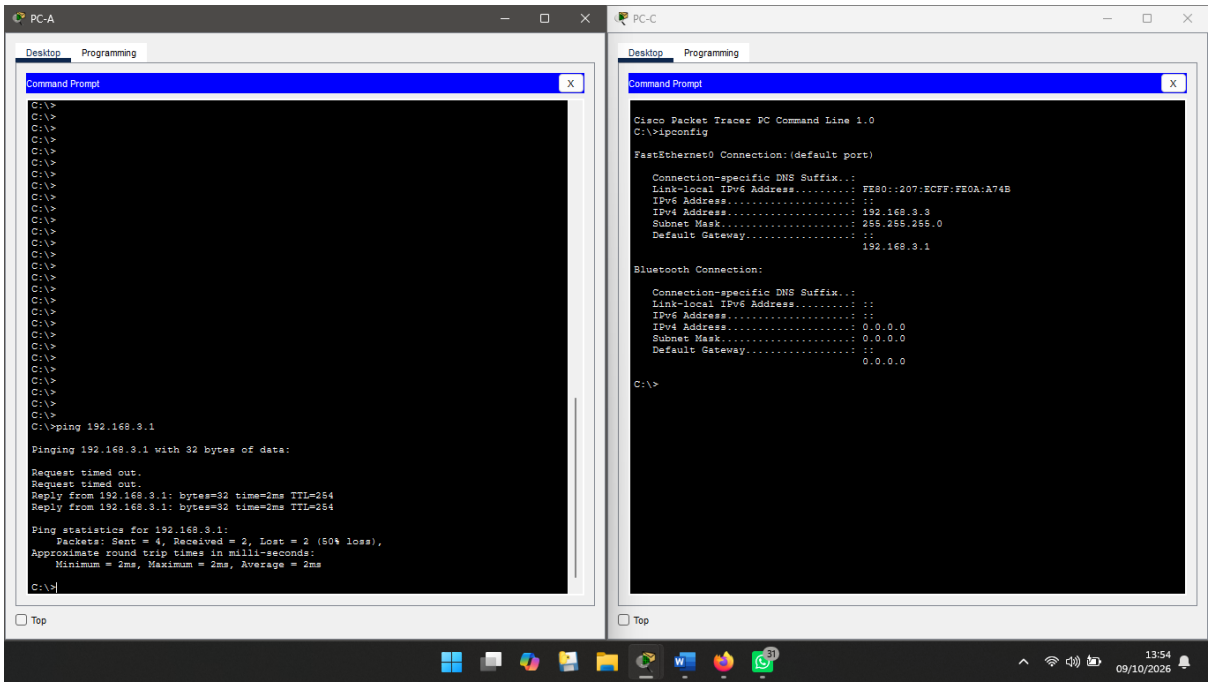
Step 1: Verify the tunnel prior to interesting traffic.

Issue the **show crypto ipsec sa** command on R1. Notice that the number of packets encapsulated, encrypted, decapsulated, and decrypted are all set to 0.



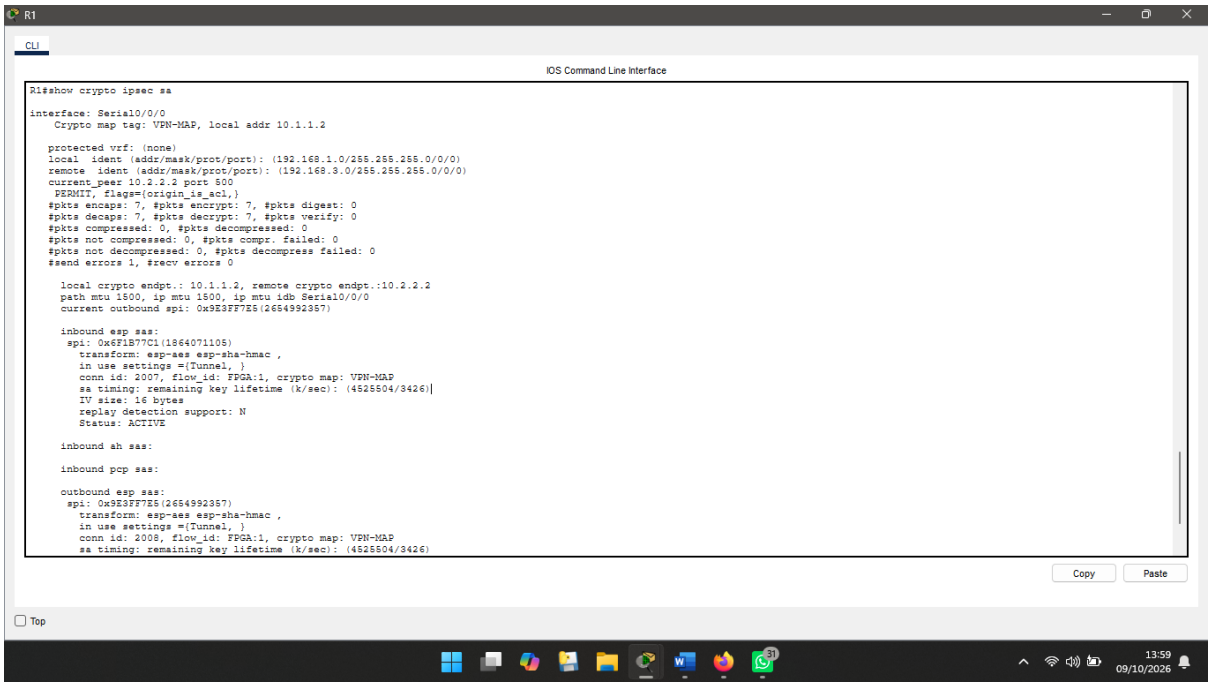
Step 2: Create interesting traffic.

Ping PC-C from PC-A.



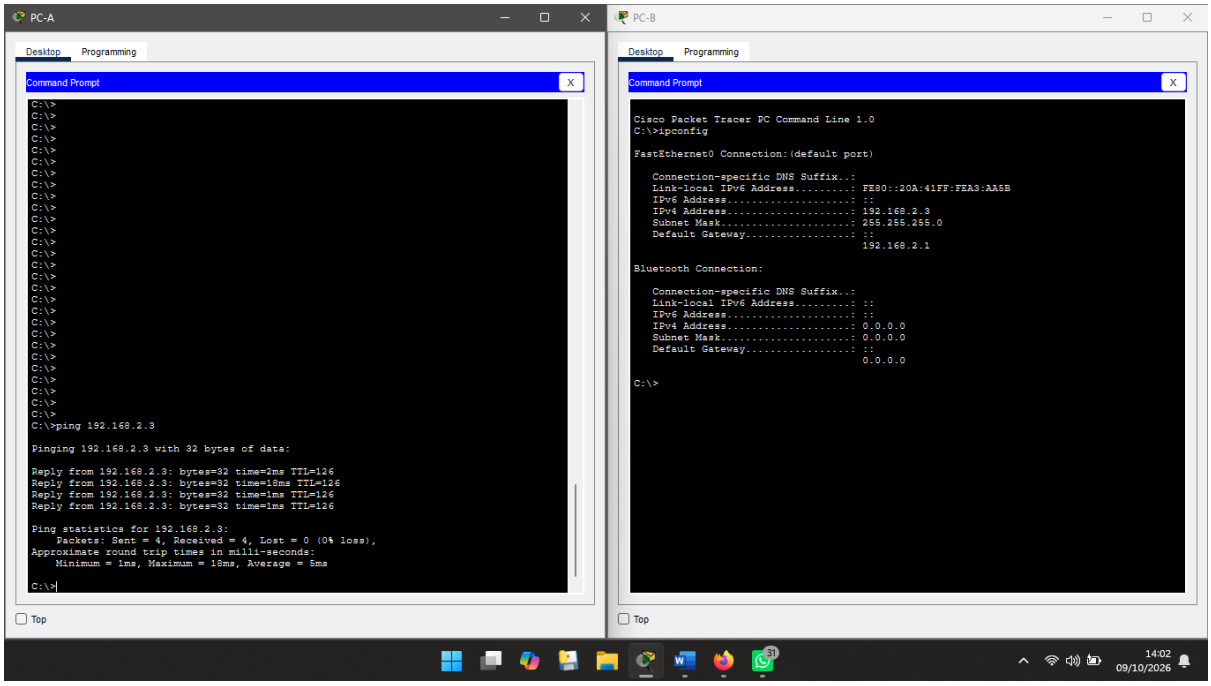
Step 3: Verify the tunnel after interesting traffic.

On R1, re-issue the **show crypto ipsec sa** command. Notice that the number of packets is more than 0, which indicates that the IPsec VPN tunnel is working.



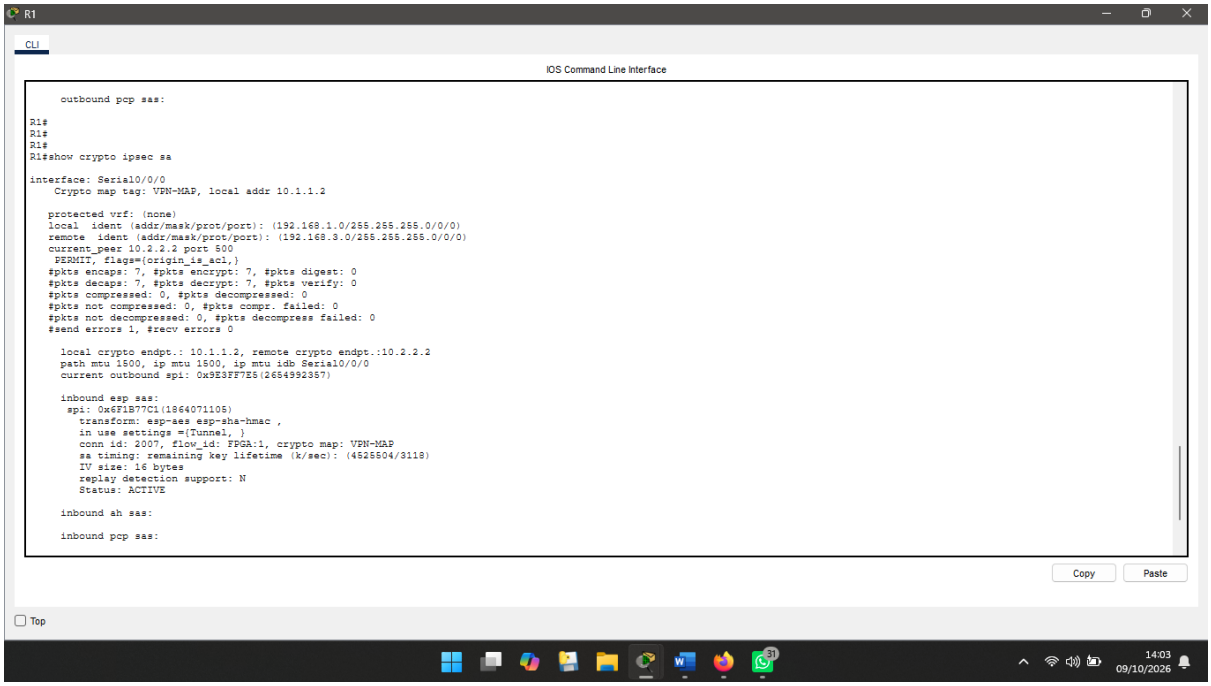
Step 4: Create uninteresting traffic.

Ping PC-B from PC-A. Note: Issuing a ping from router R1 to PC-C or R3 to PC-A is not interesting traffic.



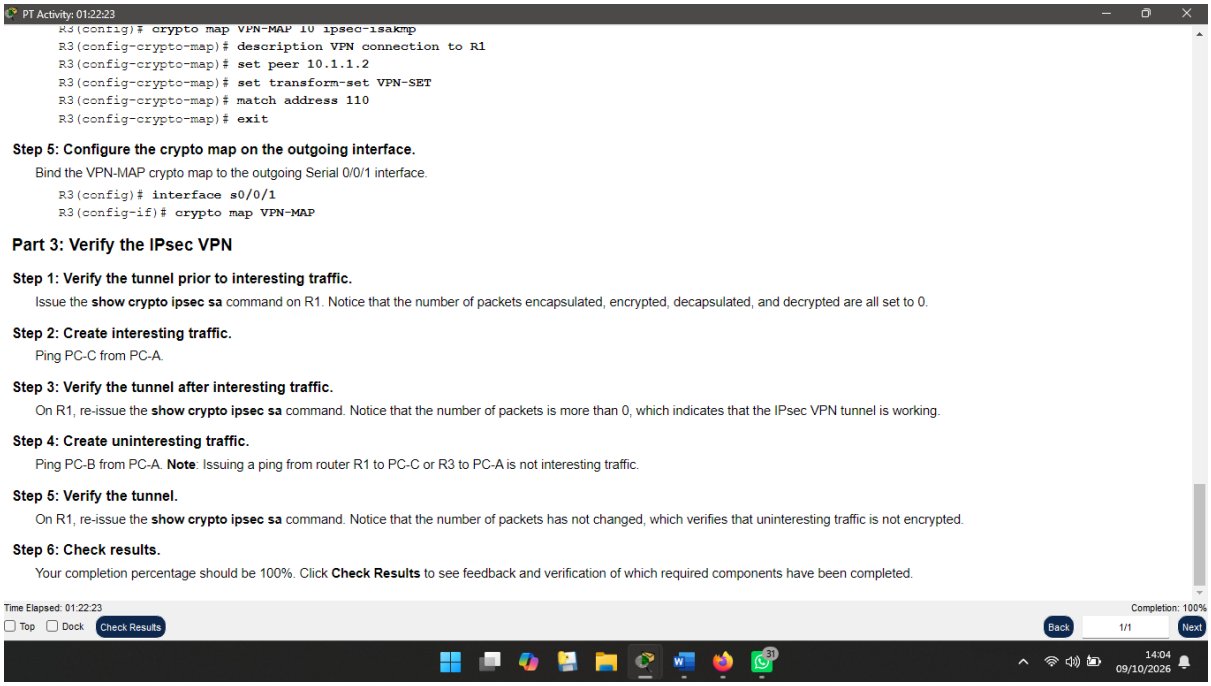
Step 5: Verify the tunnel.

On R1, re-issue the show crypto ipsec sa command. Notice that the number of packets has not changed, which verifies that uninteresting traffic is not encrypted.



Step 6: Check results.

Your completion percentage should be 100%. Click Check Results to see feedback and verification of which required components have been completed.



Conclusion

This lab showed me how a site-to-site IPsec VPN works in practice. Configuring it meant matching every parameter on both routers: the ISAKMP policy, the pre-shared key, the transform set, the crypto map, and the ACL that defines interesting traffic. A mismatch on either side would stop the tunnel from forming. I also learned that the security license must be enabled with a reload before any crypto commands will work, and that show version buries the license table below a long legal notice, which made it hard to find at first.

The verification was the most useful part. The show crypto ipsec sa counters went from 0 to 7 after pinging PC-C from PC-A, and the tunnel only came up once interesting traffic triggered it. The first ping timed out while IKE negotiated, which explained the single send error. Pinging PC-B left the counters unchanged because that traffic didn't match the ACL. R2 forwarded everything without any VPN configuration, which showed that IPsec protects traffic end to end between the two peers.